Region-level teachers in our pipeline

`-clip-mode langsplat` and the skip-vs-fill test for uncovered pixels

Radiance FYP — semantic embedding, branch `semantic/new`
intellisense08 (RTX 2080, 8 GB), 16 October 2026

Abstract

Occam's LGS run on LangSplat's features was the best design measured (25/29 objects, 1/6 absent objects firing, 14 Oct). This report covers two things. First, those region-level features are now a teacher mode of our own pipeline (`-clip-mode langsplat`), lifted with our all-pixel lift, so they can be built, cached, queried and served like every other table. Second, the open question of how to treat pixels no SAM mask covers is tested: *skip* them, as LangSplat and Occam do, or *fill* them with the scale-averaged crop feature, as our B2 variant did. With skip, our pipeline reaches **24/29 with 1/6 false positives** (dev set 5/5). That is within one query of Occam's chain, the only difference being a cluster offset on one object. **Skip vs fill differs by one query** (24 vs 23 at the default floor, 24 vs 24 at equal false positives), so filling does not explain why B2 reached only 21. The default lift mode is unchanged (A); switching it is a decision for Suhan.

1 What was built

Teacher step, `-clip-mode langsplat` (cache tag `4f4bdb1a08`)

`teachers.extract_langsplat` runs LangSplat's preprocessing for every frame as a subprocess in the `occamlgs` env, which has LangSplat's SAM fork. That means SAM ViT-H masks at levels default / s / m / l, mask NMS, and one OpenCLIP ViT-B/16 embedding per masked crop, with LangSplat's settings, which are part of the cache tag. It resumes per frame. The 720 training-view feature files from the 13–14 Oct run were moved into the cache (symlinks remain at the old path); only the 30 + 49 evaluation-split frames were extracted (37 + 31
,min). DINOv2 grids are shared with the existing teacher cache (same model and width), so L-CD stays available.

Lift, `-uncovered skip|fill` (`lift.lift_segment_views`)

Our gsplat lift, run per level $\ell \in \{s, m, l\}$. Each pixel takes the feature of the level-ℓ mask covering it; seg maps are sampled at the render's pixel centres. With $m_\ell(p) = 1$ where a mask covers p:

$$\textit{skip:}\quad f_i^{(\ell)} = \frac{\sum_{v,p} w_i(v,p)\, m_\ell(p)\, F_\ell(p)}{\sum_{v,p} w_i(v,p)\, m_\ell(p)} \qquad \text{(LangSplat / Occam's choice)}$$

$$\textit{fill:}\quad f_i^{(\ell)} = \frac{\sum_{v,p} w_i(v,p)\, \left[m_\ell F_\ell + (1-m_\ell) F_{\text{pyr}}\right](p)}{\sum_{v,p} w_i(v,p)} \qquad \text{(our B2's choice)}$$

F_{pyr} is the scale-averaged crop pyramid (teacher `d7a30d7264`). The skip denominator is one extra one-channel pass per level.

Table and query

Unchanged. The three levels are the table's three scale groups, so the query picks the level with the highest peak, as LangSplat's evaluation does. `-uncovered` is a variant key, so existing lift markers stay valid. The lift default remains `scales` (A).

Tests

Four new unit tests: skip and fill agree when every pixel is covered; skip never takes the uncovered side; fill mixes it in; seg-map sampling. Semantics suite 89 passed, Galley 116 passed.

Build commands (tables `lift_ls` and `lift_lsfill`):

```
python figs/semantic_pipeline.py --scene backroom --backend lift --clip-mode langsplat \
       --table-suffix _ls --uncovered skip
python figs/semantic_pipeline.py --scene backroom --backend lift --clip-mode langsplat \
       --table-suffix _lsfill --uncovered fill
```

2 Results

Table 1: Phase 5 sets (29 positives, 6 absent objects), query settings unchanged since Phase 1. Totals, then per scene: hits, median localisation error (m), AUROC of peak relevancy.

				backroom			flightroom		
Configuration	mode	hits	FP	hits	err	AUROC	hits	err	AUROC
Ours A (default)	L-C	22/29	2/6	9/13	0.21	0.87	13/16	0.25	0.90
	L-CD	22/29	2/6	10/13	0.25	0.87	12/16	0.25	0.92
Ours B2 (SAM ViT-B, area levels, fill)	L-C	21/29	2/6	9/13	0.19	0.87	12/16	0.20	0.94
	L-CD	21/29	1/6	10/13	0.18	0.87	11/16	0.21	0.94
Occam's full chain (centre-pixel lift)	L-C	25/29	1/6	12/13	0.27	0.92	13/16	0.30	1.00
Ours + LangSplat teachers, skip	L-C	**24**/29	1/6	12/13	0.25	0.92	12/16	0.31	1.00
	L-CD	24/29	1/6	12/13	0.25	0.92	12/16	0.33	1.00
Ours + LangSplat teachers, fill	L-C	23/29	1/6	11/13	0.24	0.92	12/16	0.30	1.00
	L-CD	23/29	1/6	11/13	0.24	0.92	12/16	0.31	1.00

Table 2: The only queries (L-C) on which these configurations differ: hit (•) or miss (○) and peak relevancy. b = backroom, f = flightroom. All other 25 queries have the same outcome in all five.

Query	Ours A	Ours B2	Occam full	LangSplat, fill	LangSplat, skip
swivel chair (b)	○ 0.621	○ 0.534	• 0.670	• 0.654	• 0.657
camera tripod (b)	○ 0.535	○ 0.508	• 0.593	○ 0.576	• 0.588
hose reel (b)	○ 0.685	○ 0.634	• 0.637	• 0.617	• 0.615
water jug (f)	• 0.634	○ 0.622	• 0.618	○ 0.631	○ 0.633
Total (of 29)	22	21	25	23	24

Table 3: Floor sweep (0.46–0.60, post-hoc on the evaluation sets): best hits with at most k absent objects firing, and hits at the default floor 0.55.

k absent objects with a candidate	1	2	3	4	5	at 0.55
Ours A (default)	19	22	23	23	24	22
Occam's full chain	25	25	25	25	25	25
no level choice	23	23	23	24	24	23
Ours + LangSplat, skip	24	24	24	24	24	24
no level choice	25	25	25	25	25	24
Ours + LangSplat, fill	24	24	24	24	24	23
no level choice	24	25	25	25	25	24

3 Findings

1. **The region teachers carry over to our pipeline.** With our all-pixel lift and skip, 24/29 with one absent object firing (only backroom's ceiling fan), AUROC 0.92 / 1.00, dev set 5/5. Like Occam's chain, it recovers *swivel chair*, *camera tripod* and *hose reel*, which A misses. The one

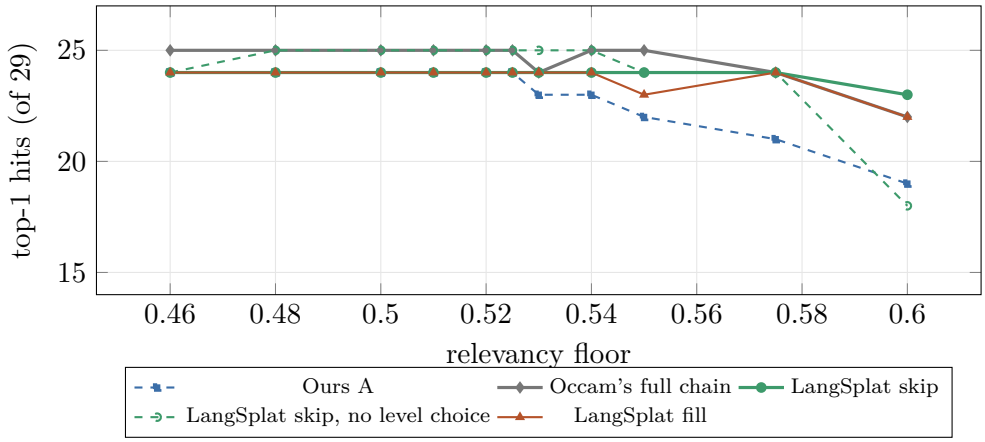


Figure 1: Hits against the relevancy floor, both scenes combined.

difference from Occam's chain is flightroom's *water jug*. Both find the jug's neighbourhood, but our cluster centroid lands outside the hit radius ("bad position"). This ±1 between the all-pixel and centre-pixel lifts matches the 13 Oct controlled comparison.

2. **Skip vs fill: one query.** The only query that differs is *camera tripod*: peak 0.588 with skip, 0.576 with fill. Filling the 6–17 % of uncovered pixels with crop features pulls a thin object's features towards its background and costs this marginal hit. At equal false positives both reach 24/29. The flat variants (no level choice) reach 25 with skip at one false positive and 24 with fill. Skip is the better choice, but the effect is small.

3. **So B2's shortfall is not the fill.** B2 reached 21/29 and misses all three region-level gains. With fill costing at most one query, the gap must come from B2's other choices: SAM ViT-B without a crop layer and with SAM's stricter default thresholds (predicted IoU 0.86 vs 0.7), levels by mask area instead of SAM's own granularities, and no LangSplat mask NMS. All three can now be tested as teacher variants of this mode.

4. **Level choice is not consistently useful.** With Occam's lift, choosing the level per query added two (25 vs 23 at one false positive). With our lift, the flat average is as good or better (25 vs 24). On 29 queries this is noise. The level choice should be kept configurable, and the default decided on a larger query set.

5. **Cost.** The lift takes 18 / 21 min per scene (three 512-channel levels plus DINO, about the same as A's lift). Occam's centre-pixel lift does the same in 30 s. Feature extraction is the expensive part: about 5 h per scene on the RTX 2080 (70 s per 1080p image), done once and cached. Tables: 2.5 / 1.8 GB, as A.

4 Decisions for Suhan and next steps

1. **Default lift mode.** `langsplat` with skip is the strongest mode in our own pipeline (24/29, 1/6; A: 22/29, 2/6), and its features are already cached for both scenes. Switching the default is one line (`default_clip_mode`). The cost is about 5 GPU-hours of extraction per new scene on the 2080.

2. **Speed the lift** by adding Occam's centre-pixel estimator as an option (30 s instead of 20 min; the 13 Oct comparison found no accuracy loss).

3. **Close the B2 gap by ablation:** SAM ViT-B vs ViT-H, crop layer on/off, area levels vs SAM levels, NMS on/off. Each needs a new SAM extraction, so they are best run on the RTX 5060 Ti or overnight.

4. **Galley:** `langsplat` tables load and query like any multi-scale table; the build should be exposed in Galley's semantics job form once the default is settled.

5 Files

Raw results: `../raw/` (Phase 5 and development-set evaluator JSONs, lift statistics including per-level pixel coverage, floor sweep, run log). Derived CSVs: `../csv/`. Tables: `tables/`, written by `../scripts/collect.py`. Code: branch `semantic/new` (`lift.py`, `teachers.py`, `semantic_pipeline.py`, `baselines/langsplat_features.py`, tests).